Advanced OOP Case Study: Smart Hospital Appointment & Billing System

Background:
A hospital management system must support patient registration, doctor management, appointment booking, billing, and revenue calculation.

Core Entities:
1. Patient
2. Doctor
3. Appointment

Requirements:
- Use OOP principles (Abstraction, Encapsulation, Polymorphism)
- Use ArrayList collections
- Implement custom exceptions
- Use Stream API for revenue calculation
- Students must complete missing method logic

```
---------------- Person.java ----------------

public abstract class Person {

    protected int id;
    protected String name;

    public Person(int id, String name) {

        if (id <= 0)
            throw new IllegalArgumentException("Invalid ID");

        // TODO: Validate name (alphabets only, min 3 chars)
        this.id = id;
        this.name = name;
    }

    public int getId() { return id; }
    public String getName() { return name; }
}

---------------- Patient.java ----------------

public class Patient extends Person {

    private List<Appointment> appointments = new ArrayList<>();

    public Patient(int id, String name) {
        super(id, name);
    }

    public void addAppointment(Appointment appointment) {
        // TODO: add to list
    }
}

---------------- Doctor.java ----------------

public class Doctor extends Person {

    private String specialization;
    private double consultationFee;
    private List<Appointment> appointments = new ArrayList<>();

    public Doctor(int id, String name,
                  String specialization,
                  double consultationFee) {

        super(id, name);

        // TODO:
        // Validate specialization not empty
        // Validate consultationFee > 0

        this.specialization = specialization;
        this.consultationFee = consultationFee;
    }

    public void bookAppointment(Appointment appointment)
            throws DoctorFullyBookedException {

        // TODO:
        // Max 5 appointments only
        // Add appointment to list
    }
}

---------------- Appointment.java ----------------

public class Appointment {

    private int appointmentId;
    private Patient patient;
    private Doctor doctor;
    private String status;
    private double billAmount;
```

```
    public Appointment(int appointmentId,
                       Patient patient,
                       Doctor doctor) {

        this.appointmentId = appointmentId;
        this.patient = patient;
        this.doctor = doctor;
        this.status = "BOOKED";
    }

    public void completeAppointment()
            throws AppointmentAlreadyCompletedException {

        // TODO:
        // If already completed throw exception
        // Change status
        // Generate bill using doctor fee
    }
}

---------------- HospitalService.java ----------------

public class HospitalService {

    private List<Patient> patients = new ArrayList<>();
    private List<Doctor> doctors = new ArrayList<>();
    private List<Appointment> appointments = new ArrayList<>();

    public void registerPatient(Patient patient) {
        patients.add(patient);
    }

    public void addDoctor(Doctor doctor) {
        doctors.add(doctor);
    }

    public Patient findPatient(int id)
            throws PatientNotFoundException {

        // TODO: Use Stream API
        return null;
    }

    public Doctor findDoctor(int id)
            throws DoctorNotFoundException {

        // TODO: Use Stream API
        return null;
    }

    public void bookAppointment(int appointmentId,
                                int patientId,
                                int doctorId)
            throws Exception {

        // TODO:
        // Find patient
        // Find doctor
        // Create appointment
        // Add to patient and doctor
        // Store in appointments list
    }

    public double calculateTotalRevenue() {

        // TODO:
        // Use Stream API
        // Sum billAmount of COMPLETED appointments
        return 0;
    }
}
```